Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	07/03/2026
Team ID	SWTID-2026-5599
Project Name	Product Inventory Management
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Authentication & Authorization**	User can register with email and password
		User can log in securely using JWT
		Only authenticated users can access dashboard
FR-2	Product Management (CRUD)	Add new product with name, price, quantity and description
		View all products in dashboard

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description

NFR-1	Usability	The application should have a simple and intuitive user interface so users can manage products easily.
NFR-2	Security	JWT authentication must be used. Passwords should be hashed and unauthorized access must be prevented.
NFR-3	Reliability	The system should provide consistent performance with minimal server errors or downtime.
NFR-4	Performance	Product data should load quickly and API responses should be returned within 2 seconds.
NFR-5	Availability	The system should remain accessible to users with minimal downtime during operation.
NFR-6	Scalability	The architecture should support increasing numbers of users and products without performance issues. Future features should be easy to integrate.